

KUSHAGRA MEHROTRA

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Education

International Institute of Information Technology, Bangalore

Aug. 2024 – Present

Master of Science by Research (Data Science)

3.53/4.0

Acropolis Institute of Technology & Research

Aug. 2019 – June 2024

Integrated Masters in Computer Applications

8.61/10.0

Experience

Yarasi Tech

May 2024 – December 2024

React Native Developer Intern

Remote

- Developed a peer-to-peer video calling app using **React Native** with **JWT authentication** and **GraphQL** backend integration.
- Added **multilingual support**, built a responsive **landing page**, and integrated **push notifications** for real-time updates.
- Utilized **Zustand** for efficient state management and ensured a seamless, scalable user experience.

Projects

Color-Guided Image Generation with ControlNet | *Python, Diffusers, ControlNet, Weights & Biases* December 2025

- Trained a ControlNet model for color-guided image generation by conditioning Stable Diffusion on color guidance maps using the `train_controlnet.py` pipeline from Diffusers.
- Designed and ran a reproducible training and evaluation workflow with Accelerate and Weights & Biases, enabling scalable experimentation and detailed metric tracking.
- Quantitatively evaluated color fidelity and semantic alignment using Mean ΔE (CIE-Lab), Block Color Error, Earth Mover's Distance (EMD), and CLIP Score, demonstrating improved adherence to target color distributions.

Fine-Tuning ViLT for Visual Question Answering | *Python, PyTorch, ViLT, LoRA, BERTScore*

March 2025

- Created a custom VQA dataset by pairing ABO product images with question-answer pairs using Gemini.
- Fine-tuned the ViLT model using LoRA under TPU memory constraints and evaluated with semantic and exact-match metrics.
- Improved BERTScore by **+30%** (0.30 $\rightarrow$ 0.64) and exact-match accuracy by **+40%** (10% $\rightarrow$ 50%) over the baseline.

High-Quality 3D Rendering with NeRF and Gaussian Splatting | *Python, COLMAP, NeRF*

December 2024

- Implemented 3D rendering pipelines to convert 2D images into high-quality 3D models using **NeRF** and **Gaussian Splatting**.
- Achieved a 90% reduction in training time with Gaussian Splatting while maintaining rendering quality, enabling rapid prototyping.

Homography Estimation Using Manual and Feature-Based Matching | *Python, OpenCV*

November 2024

- Compared **manual point matching** and **SIFT-based automated feature detection** for estimating image homography.
- Applied **RANSAC** to eliminate outliers, improving the reliability of image transformations for applications like image stitching.
- Demonstrated comparable precision between methods, showcasing their strengths for different use cases.

Technical Skills

Languages: Python, C++, Javascript

Technologies/Frameworks: Pytorch, OpenCV, NERF, Gaussian Splatting, Diffusers, WandB, WebGL, Numpy, Pandas, Matplotlib, Docker, Git, React Native

Relevant Coursework

- Machine Learning
- Visual Recognition
- Networks and Semantics
- 3D Vision
- Computer Graphics
- Generative AI for Vision